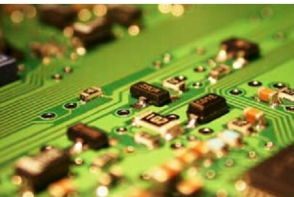


IMPROVING RELIABILITY OF A PCB DESIGN

PCB designers can improve their designs for manufacturability by following this simple checklist before freezing the design and creating numerical control data



■ SHAVINDER SINGLA

It is often observed that a circuit which functions well over a breadboard may or may not function when assembled over a PCB. Depending upon the complexity of the circuit or PCB it may take two to three days for system engineers to locate and rectify the problems.

There can be numerous reasons for such kind of failures, PCB design and fabrication being one of them. If a PCB is not designed or fabricated as per the circuit requirements, the circuit may not function or function partially. Generally, circuit designers overlook the PCB design/fabrication part so as to reduce the product's time to market or to compensate for the delay. Sometimes PCB designers or fabricators are not allowed to devote sufficient time as intended.

If the PCBs are not designed or fabricated properly, one may have to face a high level of rework/rejection of populated or unpopulated PCBs. I have also seen a lot of circuit malfunc-

tions/failures in the field, which is a result of poor design, fabrication or soldering techniques.

I personally feel that PCB designers should have a detailed knowledge of PCB fabrication as well as soldering techniques. Also, they must know the limitations of PCB manufacturers with respect to minimum drill size, minimum track width, minimum track-to-track and track-to-pad distance, tolerances, etc.

Checklist for PCB designers

PCB designers can improve their designs for manufacturability by following this checklist before freezing the design and creating numeric control (NC) data:

1. Verify that the layout is compatible with the schematic circuit diagram.
2. Ensure that the flow of the signal is maintained during PCB design.
3. Distribute the components over the entire PCB, which, in turn, will distribute weight.
4. Mark the body outline (physical size) for all the components and ensure that there is no overlapping of components.
5. Try to keep the same direction for similar components, which will reduce the errors during assembly and also the cost per assembly.
6. Check that component reference designators are not placed under the body of components, which

may create problems for a service engineer.

7. Place component reference designators in one direction followed by sufficient line width and font size.
8. Check the position of connectors with respect to mounting of the PCB as well as the wiring diagram of the equipment.
9. Check the placement of fixed-position components with respect to their origin of placement.
10. Designate all the layers of the PCB properly.
11. Place test points in the circuit.
12. Avoid placing components near the edge of the PCB, as it may create problems during automatic component placement (SMD) or during wave/reflow soldering.
13. Take extra care when selecting the package type during designing. The component body size varies from manufacturer to manufacturer for a specific component, although the pitch of the component remains the same.
14. Make sure that the position of mounting holes is as per the mechanical diagram and mounting holes are defined as non-electric unless connected to GND or any other net.
15. Provide sufficient number of mounting holes as long PCBs tend to warp if not supported properly.
16. Check the pad size vs drill size (annular ring). There should be sufficient pad area for soldering. Generally, for a single-side PCB,